DREAM REDHAT

@powered by JSNH

1.Install and Configure Ansible on the control node as follows:

* Install the required packages.

* Create a static inventory file called /home/admin/ansible/inventory as follows:

- node1 is a member of the dev host group
- node2 is member of the test host group
- node3 and node4 are member of the prod host group
- node5 is member of the balancers host group
- The prod group is a member of the webservers host group

* Create a configuration file called /home/admin/ansible/ansible.cfg

- The host inventory file is /home/admin/ansible/inventory
- The location of role used in playbooks include /home/admin/ansible/roles
- The Install Ansible Content Collection /home/admin/ansible/mycollection

✕==✕

-

2. Create a playbook for yum configuration.

As a system administrator you will need install software on the managed nodes.

Create a playbook called /home/admin/ansible/yum.yml , the user ansible create a yum repository on each of the managed nodes as follows:

Repository 1:

- The name of the repository is EX294_BASE
- The description is EX294 base software
- The base URL is http://content.example.com/rhel9.0/x86_64/dvd/BaseOS
- GPG signature checking is enabled
- The GPG Key URL is http://content.example.com/rhel9.0/x86_64/dvd/RPM-GPG-KEY-redhat-release
- The repository is enabled

Repository 2: -

- The name of the repository is EX294_STREM
- The description is EX294 base software
- The base URL is http://content.example.com/rhel9.0/x86_64/dvd/AppStream
- GPG signature checking is enabled
- The GPG Key URL is http://content.example.com/rhel9.0/x86_64/dvd/RPM-GPG-KEY-redhat-release
- The repository is enabled

✕=======================================✕

3. Install packages Create a playbook called /home/admin/ansible/packages.yml that:

- Installs the php & mariadb on hosts in the dev,test and prod host group

- Installs the RPM Development Tools package group on host in the dev host group

- Update all packages to the latest version on host in the dev host group

✕==✕

4. Ansible Content Collections.

Location of Collection inside http://utility.lab13.example.com/materails

1. install the redhat.rhel_system_roles-1.19.3.tar.gz collection
2. install the posix-1.20.2.tar.gz collection
3. install the community-general-5.5.0.tar.gz collection

✕==✕

5. Install roles using Ansible Galaxy

download and install role to /home/admin/ansible/roles from following URL:
playbook name shoud be requirement.yml

- http://server.network.example.com/materials/haproxy.tar
- The Name of the role should be balancer
- http://server.network.example.com/materials/phpinfo.tar
- The Name of the role should be phpinfo

-

✕===✕

6. Use a RHEL system role Install the RHEL System role package & create a playbook called /home/admin/ansible/timesync.yml that:

  - Runs on all managed nodes
  - Uses the timesync role
  - Configures the role to use the currently active NTP provider
  - Configure the role to use the time server 172.24.1.254
  - Configure the role to enable the iburst parameter.

✕===✕

7. Create & use a role Create a role called apache in /home/admin/ansible/roles with the following requirement:

- The httpd package is install, enabled on boot and started
- The Firewall is enabled & running with a rule to allow access the web server
- A template file index.html.j2 and is used to create the file /var/www/html/index.html with the Welcome to HOSTNAME ON IPADDRESS where HOSTNAME is the fqdn of the managed node and ip address is the ip address of managed node

* create a playbook called /home/admin/ansible/newrole.yml that uses this role as follow:

- The playbook runs on host in the webservers host group

✕===✕

8. Use roles from Ansible Galaxy Create a playbook called /home/admin/ansible/roles.yml
* The playbook contains a play that runs on host in the balances host group and uses the balancer role.

- This role configures a service to load balance web server request between hosts in the webserver host

* group. - Browsing to host in the balances host group (for example http://system5.domain1.example.com) produces the following output: Welcome to system3.domain1.example.com on 172.24.1.8

Reloading the Browser produces output from the AlterNet web server:

Welcome to system4.domain1.example.com on 172.24.1.9

* The Playbook contains a play the runs on hosts in webserver host group and user the phpinfo role. Browsing to host in the webserver host group with the URL /hell.php produces the following output : Hello PHP World from FQDN

- For example Browsing to http://system3.domain1.example.com/hello.php produces the following output: Hello PHP World from system3.domain1.example.com along with various details of the PHP configuration include the version of PHP that is installed.
@powered by JSNH

Similarly, Browsing to http://system4.doamin1.example.com/hello.php, produces the following output: Hello PHP World from system4.domain1.example.com along with various details of the PHP configuration including the version of PHP that is installed.

-

9. Generate a host file

* Download an initial template file from http://server.network.example.com/materials/hosts.j2 to /home/admin/ansible

  Complete the template so that it can he used to generate a file with a line for each host in the same format as /etc/hosts
* Create a playbook call /home/admin/host.yml that uses this template to generate the file /etc/myhosts on hosts in the dev host group.

When the playbook is run, the file /etc/myhosts on host in the dev host group should have a line for each managed hosts:

127.0.0.1 localhost localhost.localdoamin localhost4 localhost4.localdomain ::1 localhost localhost.localdoamin localhost6 localhost6.localdomain 172.24.10.6

node1.domain1.example.com node1 172.24.10.9

node2.doamin1.example.com node2 172.24.10.10

node3.doamin1.example.com node3 172.24.10.11

node4.doamin1.example.com node4 172.24.10.12

node5.doamin1.example.com node5 172.24.10.13

Note: The order in which the inventory host names appear is not important.

10. Modify file content

Create a playbook called /home/admin/ansible/issue.yml

* The playbook runs on all inventory hosts
* The playbook replaces the content of /etc/issue with a single line of text as follows:
  - On host is the dev host group, the line reads: Development
  - On host is the test host group, the line reads: Test
  - On host is the prod host group, the line reads: Production

✕======================================✕

-

11. Create a Web content directory

Create a playbook called /home/admin/ansible/webcontent.yml as follows:

* The playbook runs managed node in the dev host group

* Create the directory /webdev with the following requirement:

  - it is owner by the webdev group -it has regular permissions:

  - owner=read+write+execute,group=r+w+x,other=r+x
  - it has special permission: set group GID

* Symbolically link /webdev to /var/www/html/webdev

* Create the file /webdev/index.html with a single line of text that reads: Development

* Browsing this directory on host in the dev host group (for example http://system1.domain1.example.com/webdev/) produces the following output: Development

✕=======================================✕

12. Generate a hardware report

Create a playbook called /home/admin/ansible/hwreport.yml that produces an output file called

/root/hwreport.txt on all managed nodes with the following information: -

- Inventory hostname

- Total Memory

- BIOS version

- Size of disk device vda

- Size of disk device vdb

– Each line of the output file contains a single key=value pair.

* your playbook should Download the file from
http://server.network.example.com/materials/hwreport.empty and save it is aas /root/hwreport.txt

- Modify /root/hwreport.txt with the correct values - if a hardware item does not exist, the associated value should be set to NONE

13. Create an ansible vault to store user pasword as follows:

* The name of the vault is /home/admin/ansible/locker.yml

* The vault contains two variables with names:

- pw_developer with value Imadev

- pw_manager with value Imamgr

-

* The Password to encrypt and decrypt the vault is whenyouwishuponastar

* The password is stored in the file /home/admin/ansible/secret.txt

14. Create user accounts

* Download a list of user to be created from

http://server.network.example.com/materials/user_list.yml

* Using the password vault /home/admin/ansible/locker.yml
* Create elsewhere in this exam, create a play book called /home/admin/ansible/users.yml that create user account as following:
* User with job description of developer should be:
- create on managed node in the dev and test host group
- assigned the password from the pw_devloper variable and password should be expire in 30 days
- a member of supplementary group devops –
* User with a job description of manager should be:
create on managed node in the prod host group
- assigned the password from pw_manager variable and password should be expire in 30 days
- a member of supplementary group opsmgr *
password should use the SHA512 gas format
* Your playbook should work using the vault password file /home/admin/ansible/secret.txt create elsewhere in this exam

✕==✕

15. ReKey an Ansible vault Rekey an existing Ansible vault as follows:

- Download the Ansible vault from http://server.network.example.com/materials/salariries.yml to /home/admin/ansible/

- The current vault password is glegunge - The new vault password is migwhick

- The vault remains in an encrypted state with the new password .

✕==✕

16. Storage LVM

-

Create & use a logical Volume

Create a playbook called /home/admin/ansible/ansible/lv.yml that runs on

all managed nodes that does the following:

* Creates a logical volume with these requirement:

  - The logical Volume is Created in the research volume group

  - The logical volume name is data

  - The logical volume size is 1500 Mib

* Format the logical volume with the ext4 file system

* if the requested logical volume size cannot be created, the error message Could not create logical volume of that size should be displayed and size 800 MiB should be used instead.

* If the volume research does not exist, the error message volume group does not exist should be displayed

* Does NOT mount the logical volume in any way.

X===X

17. Use a RHEL system role

Create a playbook name selinux.yml and use system roles

Set selinux mode as enforcing in all manage node

✕==✕

18. Configure CronJob

- Create a cronjob for user natasha in all nodes, the playbook name crontab.yml and the job details are Every 2 minutes the job will execute logger "EX294 in progress"